

Chapter 16

Host Result Management

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Fig. 16.2-1: "ASTM Setup Dialog"

Protocol

☐ None

☒ ASTM

Host Configuration

Analyzer ID:

Host ID:

Automatic Operation

☒ Enable automatic downloading

☒ Enable automatic uploading

☐ Enable upload QC data

Automatic Upload Result Type

☒ All results

☐ All no needle error results

☐ All no error results

Communications

COM Port:

Baud Rate:

Data Bits:

Stop Bits:

Parity:

ASTM Delimiters

Field Delimiter:

Repeat Delimiter:

Component:

Escape Delimiter:

Current Status

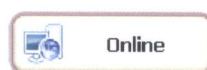
COM Port Status:

Save

Analyzer ID	This field allows assigning a unique Analyzer ID to the MAGLUMI®.
Host ID	This field allows assigning a specific name to the host (if required).
COM Port	This field specifies the serial port of the MAGLUMI® PC used for host communication. Host can be connected to any free COM port, except for COM1, which is used for the connection to the Analyzer.
Baud Rate	Specifies the Baud Rate used for transmissions between MAGLUMI® and the host any values from 4800 to 19200 can be chosen. MAGLUMI® and Host have to have matching Baud Rate speeds.
Delimiters	These fields specify the set of delimiters used for transmissions. Modification is not recommended.

16.2.2 Online icons in the MAGLUMI® software

Online icons can be found in the MAGLUMI® software.



Upload **<Online>** used for sending results from the MAGLUMI® system PC to the LIS (Host).

This icon can be used to upload results in **[Journal]** and **[Valid]**.

16.3 Types of possible Host Connections in MAGLUMI® software

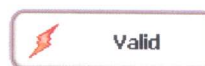
Major types of transmission between the Host and MAGLUMI® are the following:

16.3.2 How To send results to the LIS

Once the patient samples have been run and results are present, the user can send results to the Host as follows:

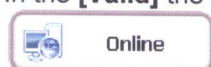
16.3.2.1 Results transmitted with validation on MAGLUMI® as manually sending of results

The results will be validated on the analyzer, and will be sent to the host manually from the **[Valid]**.



In this configuration, the **<Valid>** icon is used in the **[Report]** and all results that have been selected will be transferred to the **[Valid]**. The selected data will be deleted from the **[Journal]**.

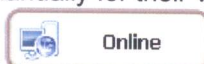
In the **[Valid]** the user must press the **<Online>** icon



(Uploading) to send all results to the Host. The transfer is completed when the status reads **[Finished]**.

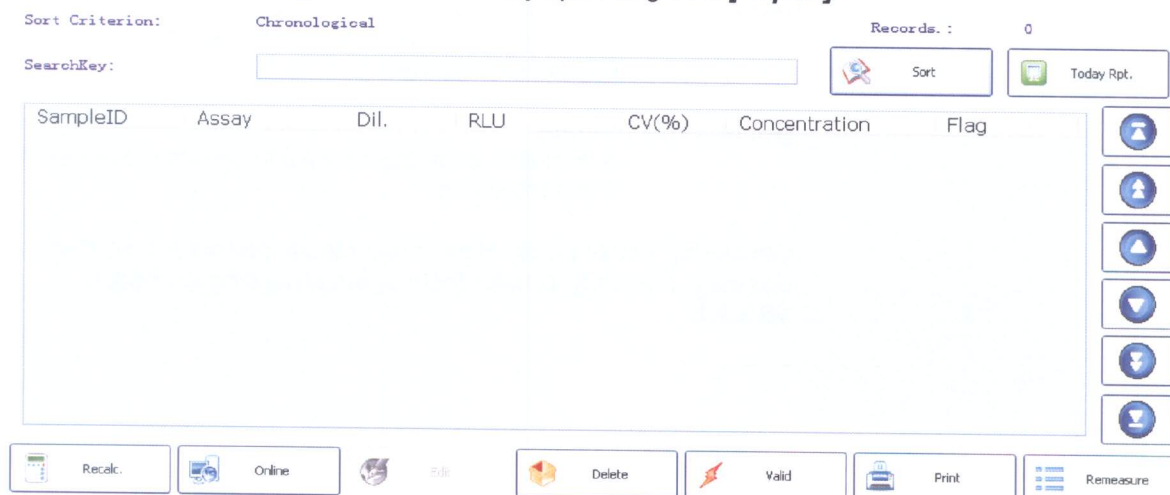
16.3.2.2 Results transmitted without validation on MAGLUMI® as manually sending of results

The results can not be validated on the analyzer but will instead be sent to the host manually for their validations. In this configuration,



an **<Online>** icon (Uploading) appears in the **[Journal]** (as pictured in *fig. 16.3.2-1 manually uploading from [Report]*).

Fig. 16.3.2-1 manually uploading from **[Report]**



16.3.2.3 Results transmitted without validation on MAGLUMI® as an automatic sending of results

The results can not be validated on the analyzer but will instead be sent to the host automatically for their validations.

16.4 MAGLUMI® software communication format

MAGLUMI® software use ASTM E1394 protocol for communication. Directory \online\log records all the communications between the software and LIS.

The content of the communication between MAGLUMI® software and LIS is as following:

16.4.1 Inquiry Reagent Information

Once the sample rack is inserted in to the sample area, after the barcode reader reads the barcode labeled on the sample tube, MAGLUMI® software will inquiry the LIS host for the assay need to be performed for the sample.

Communication content:

```
--><ENQ>
<--<ACK>
--><STX>
<--<ACK>
-->H|\^&||PSWD|Maglumi 600||||Lis||P|E1394-97|20100323<CR>
Q|1|^1234567||ALL||||||O<CR>
L|1|N<CR>
<--<ACK>
--><ETX>
<--<ACK>
--><EOT>
<--<ACK>
```

Symbol meaning:

Letter	means	ASCII
-->	send	
<--	receive	
<ENQ>	enquiry	0x05
<ACK>	answer confirm	0x06
<STX>	text start	0x02
<ETX>	text ends	0x03
<CR>	home	0x0D
<EOT>	transmitting end	0x04

Here:

```
H|\^&||PSWD|Maglumi 600||||Lis||P|E1394-97|20100323<CR>
Q|1|^1234567||ALL||||||O<CR>
L|1|N<CR>
```

This is a message which ASTM E1397 enquiry the assay needs to be performed for the sample. Here we take sample ID 1234567 for example.

16.4.3 Sending test results

```

--><ENQ>
<--<ACK>
--><STX>
<--<ACK>
-->H|\^&||PSWD|Maglumi 600||||Lis||P|E1394-97|20100326<CR>
P|1<CR>
O|1|1234567||^^^CYFRA211<CR>
R|1|^^^CYFRA211|0||0 to 7|N|||||20100326172956<CR>
L|1|N<CR>
<--<ACK>
--><ETX>
<--<ACK>
--><EOT>
<--<ACK>

```

Letter meaning as above,

Here:

```

H|\^&||PSWD|Maglumi 600||||Lis||P|E1394-97|20100326<CR>
P|1<CR>
O|1|1234567||^^^CYFRA211<CR>
R|1|^^^CYFRA211|0||0 to 7|N|||||20100326172956<CR>
L|1|N<CR>

```

This is a message which ASTM E1394 sending the test results to LIS.
Here sample ID 1234567 performed assay CYFRA211 test result is sent to LIS.